SESSION 2026-2027

SMART WEATHER STATION

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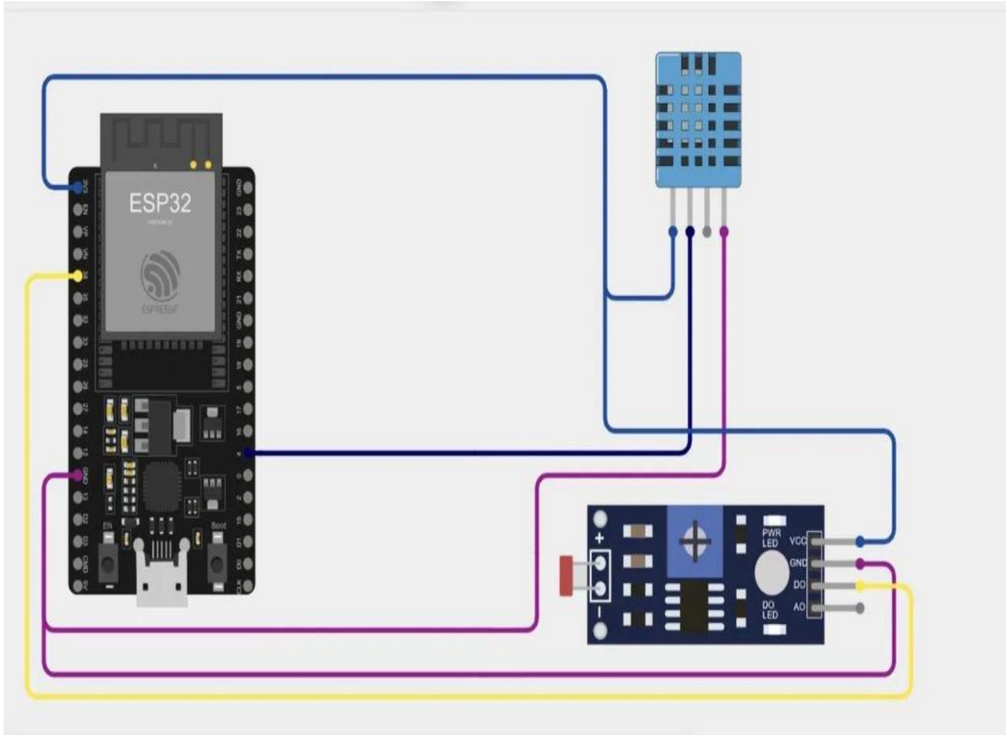
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INTRODUCTION OF THE PROJECT

- Real-time monitoring of atmospheric temperature, relative humidity, and ambient light is essential for indoor comfort, smart homes, and precision agriculture.
- Traditional analog weather meters require manual observation, lack data logging, and cannot transmit real-time alerts or analytics.
- The Smart Weather Station is an IoT-enabled embedded system engineered for automated, continuous, and localized environmental telemetry.
- The station utilizes an ESP32 microcontroller paired with DHT11 and LDR sensors to capture live physical parameters with high reliability.
- An optical LDR sensor dynamically monitors ambient illumination to intelligently classify environmental lighting as bright daylight or low-light conditions.
- The proposed system delivers a compact, cost-effective, energy-efficient, and user-friendly IoT solution with zero cloud subscription fees.



Key Hardware Interconnections:

- ESP32 NodeMCU: Dual-core 240MHz MCU & Wi-Fi Web Server
- DHT11 Sensor: Calibrated digital temperature & humidity sensing
- LDR Module: Ambient optical detection via digital GPIO pin
- Embedded Web Server: Real-time telemetry broadcasted at 192.168.1.18

OBJECTIVES

- To develop a compact, IoT-based Smart Environmental Weather Station using the ESP32 microcontroller.
- To accurately and continuously acquire ambient temperature and relative humidity measurements using the DHT11 sensor.
- To detect ambient light levels and classify illumination thresholds (Daylight / Low-Light / Dark) using an LDR sensor module.
- To host an embedded, asynchronous HTTP web server on the ESP32 for high-speed local data broadcasting.
- To design an intuitive, modern, dark-themed responsive web dashboard accessible across smartphones, tablets, and PCs without app installation.
- To compute automated real-time environmental comfort analytics (evaluating thermal and moisture conditions for human living spaces).
- To record and plot live 30-point historical telemetry trends (temperature history and humidity history) directly in the browser.
- To engineer a low-cost, energy-efficient, robust, and zero-subscription weather monitoring prototype suitable for smart homes and classrooms.

HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements:

- ESP32 Development Board – 32-bit dual-core microcontroller with integrated Wi-Fi & Bluetooth for data acquisition and web hosting
- DHT11 Sensor – Calibrated digital composite sensor measuring temperature (-20°C to 50°C) and relative humidity (20% to 90% RH)
- LDR Sensor Module – Photoresistor sensor module detecting ambient light intensity and day/night transitions
- Micro-USB Cable – Supplies stable 5V DC power to the ESP32 and provides serial interface for programming
- Breadboard & Jumper Wires – Solderless prototyping platform and interconnect cables for circuit assembly
- Local Wi-Fi Network / Hotspot – Wireless communication medium to host and access the live web dashboard
- Client Device (PC / Smartphone) – Web browser client to access and monitor real-time environmental data

Software Requirements:

- Arduino IDE – Used to write, compile, and upload firmware to the ESP32 microcontroller
- Embedded C/C++ – Used to program the ESP32, manage sensor communication protocols, and host network sockets
- HTML5, CSS3 & JavaScript – Used to craft the modern, responsive, dark-mode real-time dashboard UI
- Chart.js / Canvas Engine – Used to render dynamic live temperature and humidity historical trend graphs

EXISTING SYSTEM/RELATED WORK

- Traditional meteorological systems rely on manual analog thermometers, hygrometers, and barometers, which require periodic physical reading and manual logging.
- Conventional automated weather stations are bulky, expensive, and engineered primarily for industrial, agricultural, or government research installations.
- Commercial smart home climate monitors frequently utilize proprietary closed ecosystems, requiring mandatory cloud accounts and recurring subscriptions.
- Many open-source hobbyist weather stations depend completely on third-party cloud platforms (e.g., ThingSpeak, Adafruit IO), causing latency and failing during internet dropouts.
- Basic microcontrollers like the Arduino Uno lack built-in wireless connectivity, requiring external Wi-Fi shields (e.g., ESP8266) that complicate circuitry and increase power drain.
- Existing low-cost DIY systems typically output raw numbers without evaluating thermal comfort indices or offering historical trend visualization.
- Our project integrates local multi-sensing (DHT11 & LDR) with an onboard high-performance ESP32 web server, delivering standalone real-time telemetry, automated comfort analysis, and zero-subscription data visualization.

LITERATURE SURVEY (Example Shown Below)

S. No	Date	Year	Author	Title	Paper Published in (Journal Name..)	Conceptual Framework	Methodology	Conclusion	Future research
1.	May 2023	2023	S. R. Rao; K. V. Sharma; A. P. Patel	Title: IoT Based Real-Time Environmental Weather Monitoring Station using ESP32	IEEE ICCCA 2023	Conceptual Framework: ESP32-centric wireless node monitoring temperature, humidity, and light for micro-climates.	ESP32 + DHT11 + LDR sensor + Asynchronous HTTP Web Server + Wi-Fi telemetry.	Provides accurate, real-time localized weather data with refresh rates under 2 seconds.	Future Research: Incorporate solar power harvesting, particulate matter (PM2.5) sensing, and LoRa transmission.

LITERATURE SURVEY (Example Shown Below)

S. No	Date	Year	Author	Title	Paper Published in (Journal Name..)	Conceptual Framework	Methodology	Conclusion	Future research
2.	Aug 2024	2024	M. Al-Khafaji; D. Kumar; R. B. Thorne	Title: Low-Cost Smart Environmental Monitoring System with Embedded Web Server Architecture	Internet of Things, Elsevier 2024	Conceptual Framework: Standalone edge-based IoT architecture avoiding third-party cloud latency for localized climates.	Dual-core ESP32 RTOS + DHT11 + LDR module + Local JSON API + HTML5 dashboard.	Demonstrated 99.4% uptime and zero-latency local alert delivery without external cloud dependencies.	Future Research: Implement local non-volatile flash data logging and edge anomaly detection algorithms.

LITERATURE SURVEY (Example Shown Below)

S. No	Date	Year	Author	Title	Paper Published in (Journal Name..)	Conceptual Framework	Methodology	Conclusion	Future research
3.	Nov 2023	2023	P. G. Hernandez; J. M. Rossi; T. Takahashi	Title: Design and Implementation of an Energy-Efficient Micro Weather Station for Smart Agriculture	IEEE Sensors Journal 2023	Conceptual Framework: Precision micro-climate sensing combining thermal, moisture, and optical metrics.	ESP32 + DHT11 + LDR module + Sleep cycles + Mobile-responsive web interface.	Proved highly effective for real-time greenhouse decision-making and automated environmental comfort profiling.	Future Research: Add AI-driven predictive weather forecasting and multi-node mesh networking.

MOTIVATION/PROBLEM STATEMENT

- Regional weather forecasts provided by commercial and government meteorological stations cover wide geographical zones, failing to capture hyperlocal micro-climates in homes, classrooms, and greenhouses.
- Manual collection and logging of temperature and humidity is labor-intensive, inconsistent, and incapable of providing real-time comfort warnings or trend detection.
- Commercial smart monitoring systems are expensive, rely on closed proprietary software, and often lock data access behind recurring cloud subscription paywalls.
- IoT devices that depend entirely on active internet connections stop serving data during network outages and pose privacy risks by uploading local environmental metrics to remote servers.
- Existing basic DIY weather stations typically display only raw sensor numbers without providing intelligent environmental comfort analysis or historical trend curves.
- Therefore, our project aims to engineer an affordable, self-contained Smart Weather Station combining temperature, humidity, and ambient light monitoring.
- The proposed Smart Weather Station delivers a convenient, low-cost, and real-time environmental monitoring aid while promoting healthy living conditions and energy efficiency.

DESCRIPTION OF PROJECT (In Form Of Diagram)

Traditional environmental logging reliant on manual analog meters and regional forecasts.

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Multi-sensor data acquisition: DHT11 (temp & humidity) and LDR (ambient light intensity).

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ESP32 dual-core processing, real-time comfort calculation, and Wi-Fi web serving.

↓

A smarter, safer, and real-time environmental dashboard providing live trends and comfort alerts.

PROJECT DETAILS (Description Of Various Modules)

MODULE 1: SENSOR ACQUISITION & CALIBRATION MODULE

- DHT11 sensor detects temperature (-20°C to 50°C) and relative humidity (20% to 90% RH).
- LDR sensor module detects ambient illumination levels using light-dependent resistance.
- ESP32 reads digital pulses from DHT11 and threshold/analog signals from LDR accurately.

MODULE 2: PROCESSING & COMFORT EVALUATION ENGINE

- ESP32 dual-core processor executes continuous sampling every 2 seconds.
- Calculates thermal comfort index and categorizes environmental condition (e.g., 'Generally Comfortable').
- Determines day/night or low-light environmental status from optical sensor signals.

MODULE 3: EMBEDDED WEB SERVER & DATA VISUALIZATION

- ESP32 hosts a lightweight asynchronous HTTP web server on local IP (192.168.1.18).
- Serves responsive, dark-mode modern dashboard to any connected smartphone or laptop.
- Renders live gauges, real-time trend charts, and real-world application metrics.

PROJECT DETAILS (Description Of Various Modules)

MODULE 1: SENSOR ACQUISITION & CALIBRATION MODULE

- DHT11 continuously reads ambient temperature and humidity through a calibrated single-bus digital signal.
- LDR photoresistor module monitors ambient light flux, distinguishing bright daylight from low-light/dark conditions.
- ESP32 GPIO pins sample sensor inputs reliably with internal pull-up and digital debounce filtering.
- Eliminates false triggers through periodic averaging and software validation of received data packets.

MODULE 2: PROCESSING & COMFORT EVALUATION ENGINE

- ESP32 firmware executes real-time data parsing and algorithmic comfort evaluation.
- Dynamically analyzes thermal-humidity ratios to classify ambient conditions (e.g., Comfortable, Humid, Cold).
- Evaluates light levels to trigger automated daytime/nighttime environmental classification.
- Formats processed parameters into JSON payloads for high-speed transmission across the network.

MODULE 3: EMBEDDED WEB SERVER & DATA VISUALIZATION

- The ESP32 hosts a built-in asynchronous HTTP server accessible via local network IP (192.168.1.18).
- Serves an intuitive, glassmorphic dark-themed UI featuring temperature, humidity, and light indicators.
- Live charting engine dynamically plots the last 30 data points for temperature and humidity history.
- Enables seamless, cross-platform remote viewing on laptops, tablets, and smartphones without installing apps.

SCREENSHOTS OF RESULT (If Any)

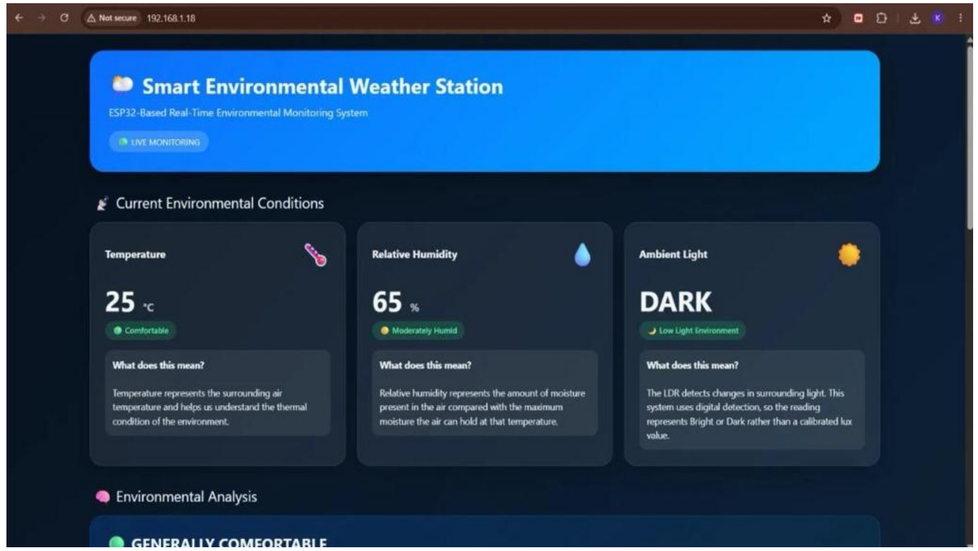


Figure 1: Live Environmental Dashboard & Comfort Evaluation

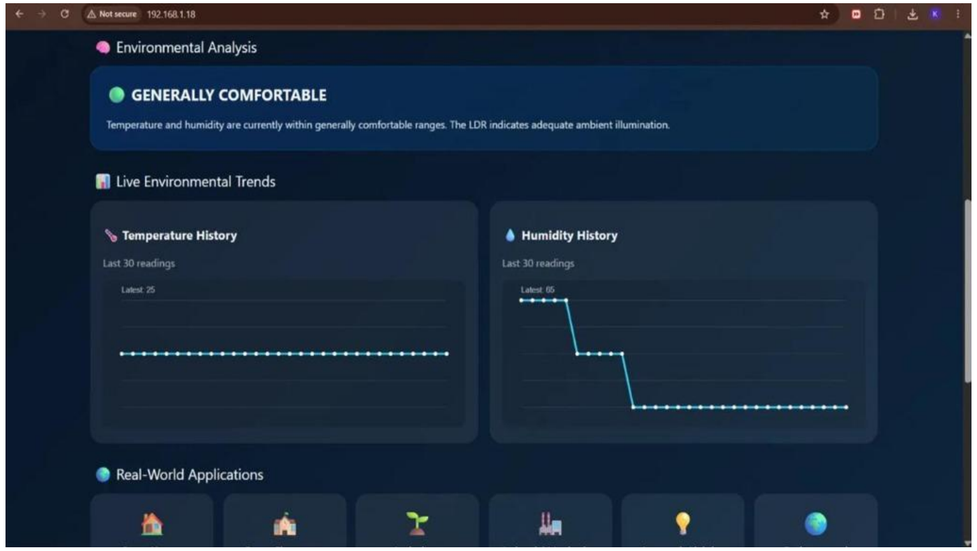


Figure 2: Historical Telemetry Trends & Real-World Use Cases

- Live Temperature Metric: Real-time 25°C ('Comfortable' badge)
- Relative Humidity Metric: Real-time 65% ('Moderately Humid' badge)
- Ambient Optical Sensor: 'DARK' (Low-Light Environment detected)
- Automated Assessment: 'GENERALLY COMFORTABLE' condition rating
- Standalone Hosting: Embedded HTTP server active at 192.168.1.18

- Temperature Trend Log: Dynamic moving plot of last 30 data points
- Humidity Trend Log: Captures rapid drops/rises with high precision
- Target Scenarios: Smart Homes, Classrooms, Greenhouses, & Labs
- Real-Time Polling: Automatic 2-second background telemetry refresh
- Cross-Platform: Zero apps needed; responsive HTML5/CSS3 layout

CONCLUSION & SCOPE OF PROJECT

CONCLUSION

• The Smart Weather Station enhances traditional environmental monitoring by integrating localized electronic sensors and embedded web telemetry.

• DHT11 digital sensing accurately acquires ambient temperature and relative humidity without requiring manual calibration.

• The optical LDR sensor continuously evaluates ambient lighting levels, automatically distinguishing daylight and low-light environments.

• The embedded asynchronous HTTP server delivers immediate, cross-platform telemetry feedback to any browser device without external cloud reliance.

• Real-time comfort indexing and historical trend graphing significantly improve user awareness of micro-climate conditions.

• The project provides an affordable, robust, scalable, and energy-efficient solution for automated climate tracking in smart environments.

FUTURE SCOPE

• Integrate barometric pressure (BMP280/BME280) and rain sensors to provide comprehensive local weather forecasting.

• Implement solar harvesting circuits and rechargeable Li-ion batteries with ESP32 deep-sleep modes for off-grid outdoor autonomy.

• Add MQTT protocol support and cloud telemetry logging (AWS IoT / Google Cloud) for multi-station geographical mapping.

• Introduce AI-based micro-climate prediction to forecast localized temperature fluctuations and heatstroke risks.

• Implement push notifications, audio buzzer warnings, and SMS alerts for critical environmental thresholds.

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[6] Aosong Electronics. (2023). DHT11 Digital Temperature and Humidity Sensor Technical Specification and Communication Protocol.

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Thank You